

Lab 3 Architecture Justification

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Problem Statement 47 | Domain & SSL Certificate Expiry Alert System

Architecture selection

We chose Layered Architecture for the Domain & SSL Certificate Expiry Alert System. The presentation layer contains the dashboard; the application layer manages endpoints, daily audits and alert escalation; the infrastructure layer provides TLS, WHOIS, notification and database adapters.

Comparison of architectural styles

Layered architecture offers clear responsibilities and straightforward testing, although scaling one module independently is harder. Microservices allow independent probe scaling and deployment, but introduce network failures, deployment overhead and distributed alert-state consistency. Client-server centralizes access and data, but a single server can become a bottleneck and it does not itself define internal module boundaries. Layered architecture best fits this project; its dashboard can still communicate with the application through a client-server API.

Two scenario specific reasons

1. Replaceable integrations: TLS certificate checks and WHOIS lookups use separate interfaces. A change in WHOIS parsing, timeout handling or the notification provider can be confined to an adapter without rewriting the dashboard or alert rules.
2. Consistent audit and escalation state: a shared repository records endpoints, results, alerts and acknowledgements. The alert component owns the 30/15/3-day rules and configured escalation ladder. Transactional state checks and idempotent dispatch prevent duplicate processing and stop subsequent escalation after acknowledgement.

Security advantage

The application enforces role-based access before endpoint changes, history access or acknowledgements. The dashboard has no direct database or secret access. Infrastructure adapters use least-privilege credentials, encrypted secret storage and TLS for application and notification APIs; WHOIS responses are treated as untrusted external data.

Performance benefit

A bounded asynchronous worker pool overlaps slow TLS and WHOIS calls while the dashboard reads stored results. Timeouts, limited retries and provider rate limits keep one failed endpoint from blocking the audit. This supports the Lab 1 target of 1,000 endpoints in under 3 minutes; a workload benchmark is still required to verify that target.